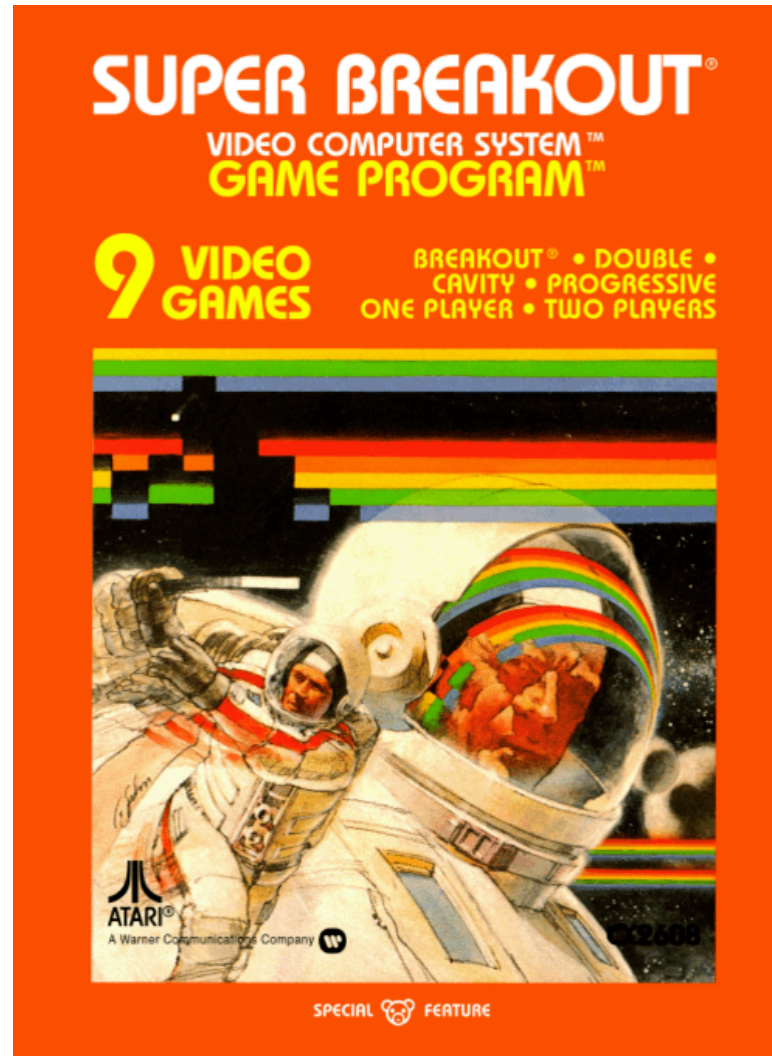


RL for POMDPs

4 Approaches

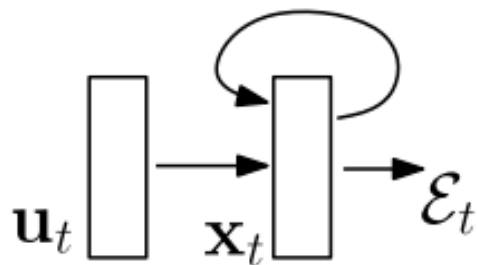
Approach 1: k-Markov



Approach 2: Recurrent Neural Network

$$\mathbf{x}_t = F(\mathbf{x}_{t-1}, \mathbf{u}_t, \theta)$$

$$\mathbf{x}_t = \mathbf{W}_{rec}\sigma(\mathbf{x}_{t-1}) + \mathbf{W}_{in}\mathbf{u}_t + \mathbf{b}$$



Input: u_t

State/Output: x_t

Cost: $\mathcal{E}_t$

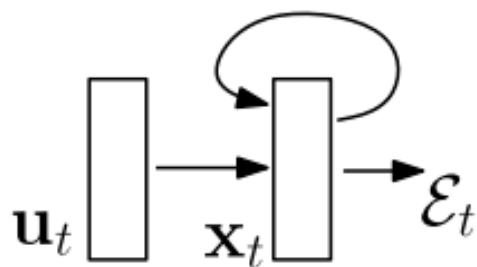
Images: Pascanu 2013

e.g. Alpha Star

Approach 2: Recurrent Neural Network

$$\mathbf{x}_t = F(\mathbf{x}_{t-1}, \mathbf{u}_t, \theta)$$

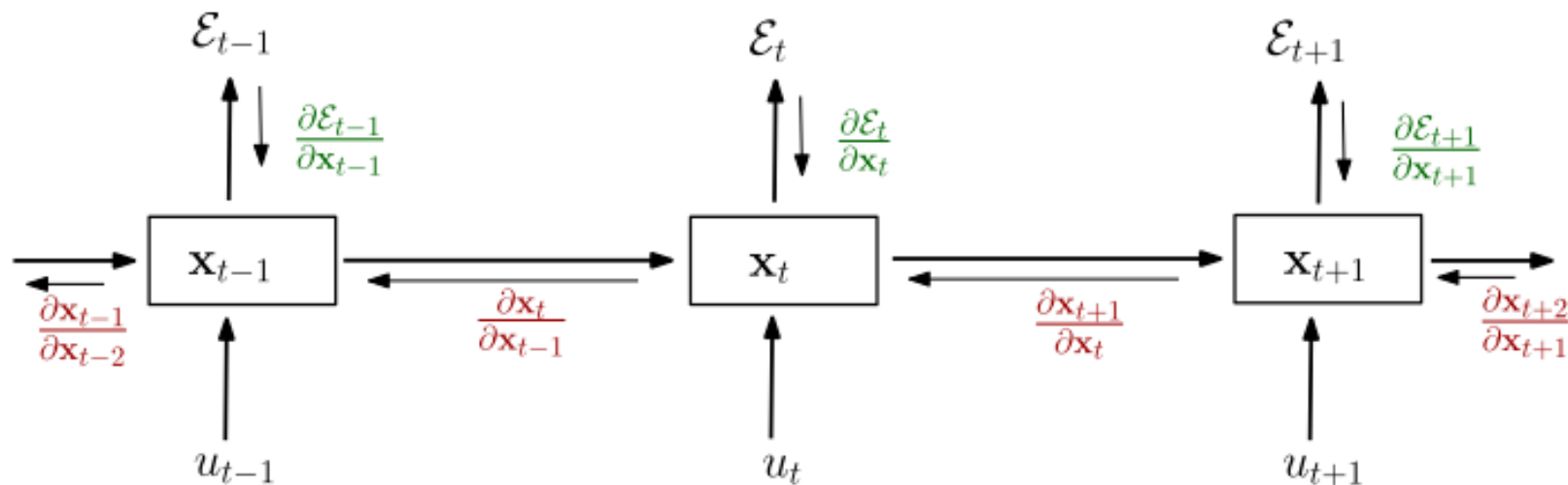
$$\mathbf{x}_t = \mathbf{W}_{rec}\sigma(\mathbf{x}_{t-1}) + \mathbf{W}_{in}\mathbf{u}_t + \mathbf{b}$$



Input: u_t

State/Output: x_t

Cost: $\mathcal{E}_t$



Images: Pascanu 2013

e.g. Alpha Star

Approach 2: RNN (e.g. LSTM)

Input: x_t

Output: h_t

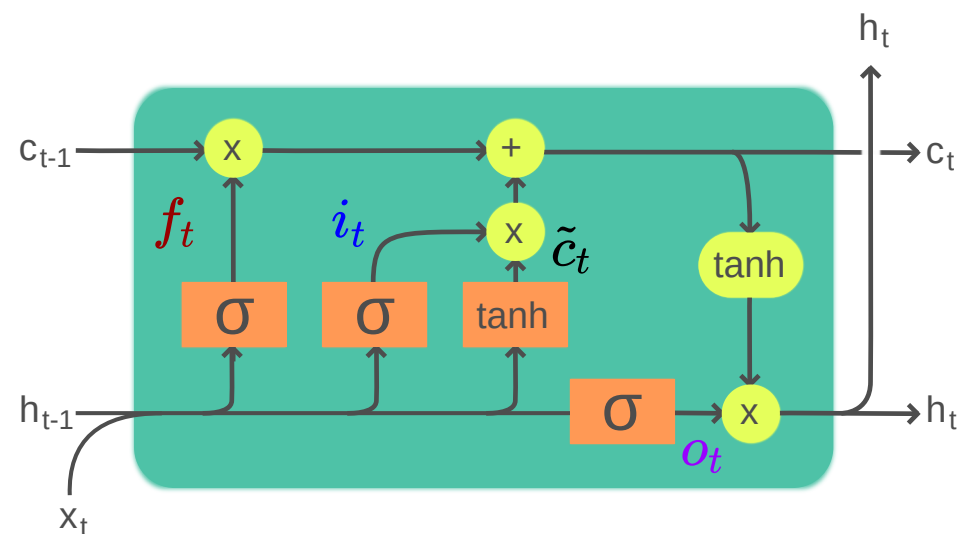
Cell state: c_t

Approach 2: RNN (e.g. LSTM)

Input: x_t

Output: h_t

Cell state: c_t



Legend:

Layer



Componentwise



Copy



Concatenate



By Guillaume Chevalier - File:The_LSTM_Cell.svg, CC BY-SA 4.0, <https://commons.wikimedia.org/w/index.php?curid=109362147>

Approach 2: RNN (e.g. LSTM)

Input: x_t

Output: h_t

Cell state: c_t

$$f_t = \sigma_g(W_f x_t + U_f h_{t-1} + b_f)$$

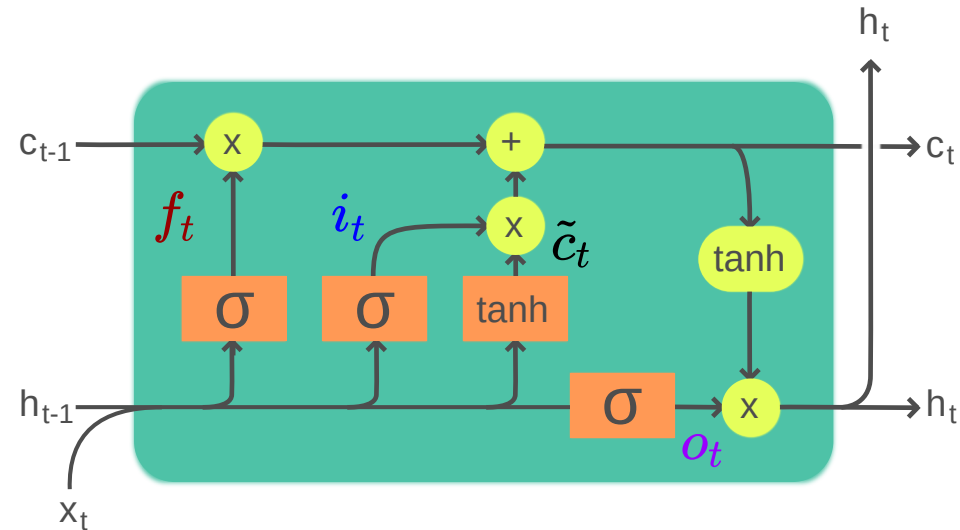
$$i_t = \sigma_g(W_i x_t + U_i h_{t-1} + b_i)$$

$$o_t = \sigma_g(W_o x_t + U_o h_{t-1} + b_o)$$

$$\tilde{c}_t = \sigma_c(W_c x_t + U_c h_{t-1} + b_c)$$

$$c_t = f_t \odot c_{t-1} + i_t \odot \tilde{c}_t$$

$$h_t = o_t \odot \sigma_h(c_t)$$



Legend:

Layer

Componentwise

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Approach 2: RNN (e.g. LSTM)

Input: x_t

Output: h_t

Cell state: c_t

$$f_t = \sigma_g(W_f x_t + U_f h_{t-1} + b_f) \quad \text{Forget gate}$$

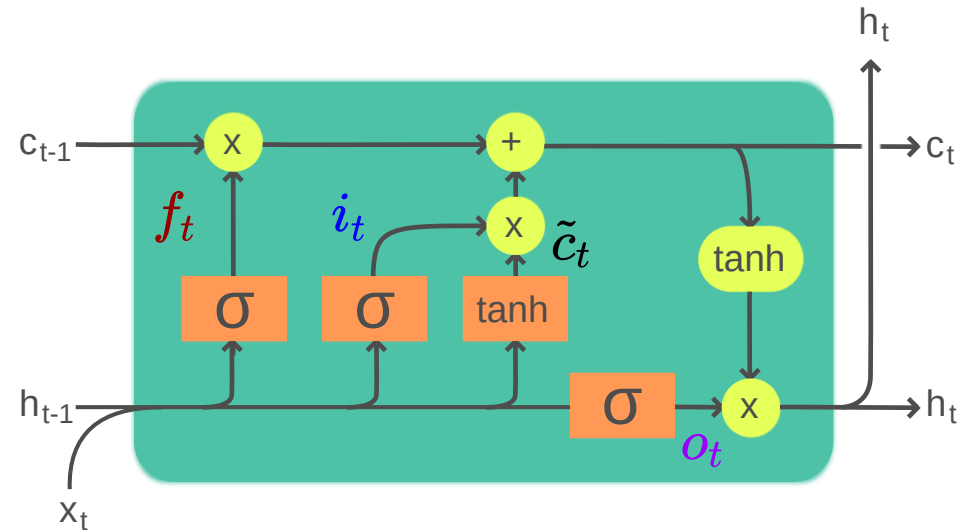
$$i_t = \sigma_g(W_i x_t + U_i h_{t-1} + b_i)$$

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$$c_t = f_t \odot c_{t-1} + i_t \odot \tilde{c}_t$$

$$h_t = o_t \odot \sigma_h(c_t)$$



Legend:

Layer

Componentwise

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Approach 2: RNN (e.g. LSTM)

Input: x_t

Output: h_t

Cell state: c_t

$$f_t = \sigma_g(W_f x_t + U_f h_{t-1} + b_f) \quad \text{Forget gate}$$

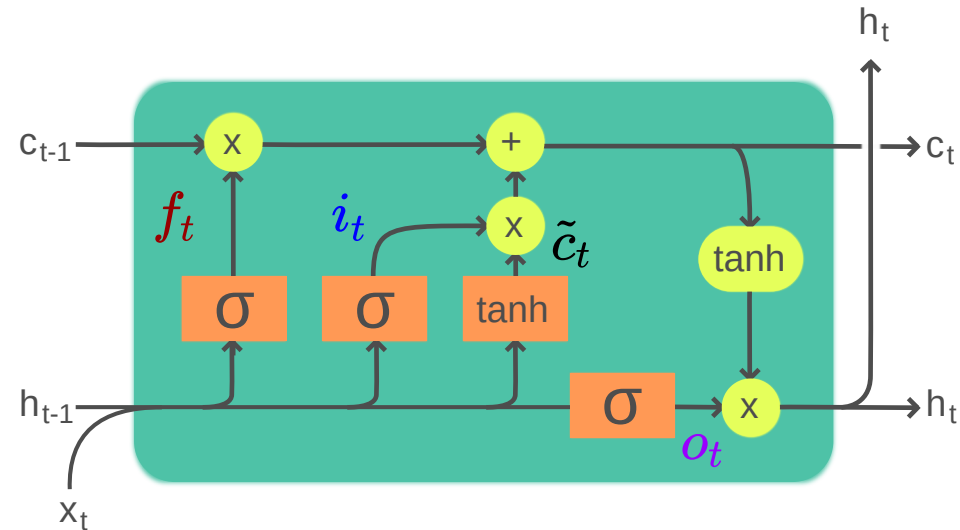
$$i_t = \sigma_g(W_i x_t + U_i h_{t-1} + b_i) \quad \text{Input Gate}$$

$$o_t = \sigma_g(W_o x_t + U_o h_{t-1} + b_o)$$

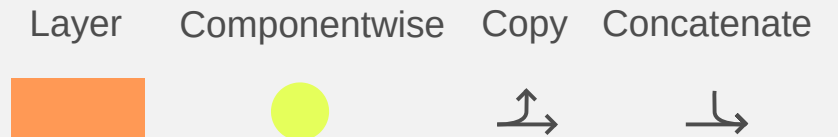
$$\tilde{c}_t = \sigma_c(W_c x_t + U_c h_{t-1} + b_c)$$

$$c_t = f_t \odot c_{t-1} + i_t \odot \tilde{c}_t$$

$$h_t = o_t \odot \sigma_h(c_t)$$



Legend:



By Guillaume Chevalier - File:The_LSTM_Cell.svg, CC BY-SA 4.0, <https://commons.wikimedia.org/w/index.php?curid=109362147>

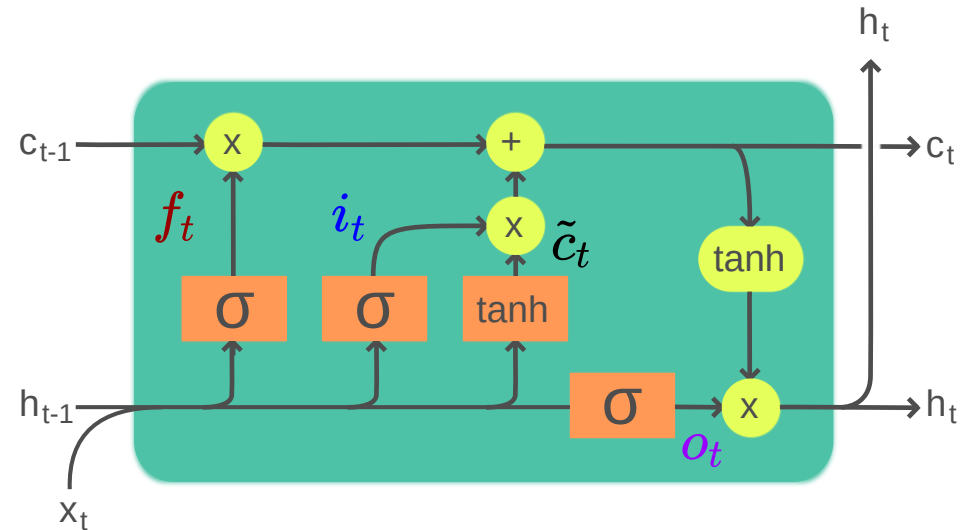
Approach 2: RNN (e.g. LSTM)

Input: x_t

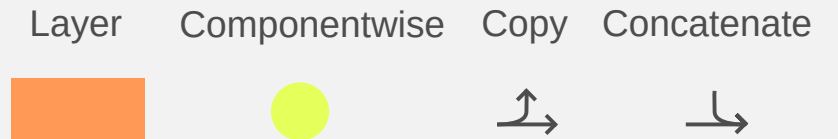
Output: h_t

Cell state: c_t

$$\begin{aligned} f_t &= \sigma_g(W_f x_t + U_f h_{t-1} + b_f) && \text{Forget gate} \\ i_t &= \sigma_g(W_i x_t + U_i h_{t-1} + b_i) && \text{Input Gate} \\ o_t &= \sigma_g(W_o x_t + U_o h_{t-1} + b_o) && \text{Output Gate} \\ \tilde{c}_t &= \sigma_c(W_c x_t + U_c h_{t-1} + b_c) \\ c_t &= f_t \odot c_{t-1} + i_t \odot \tilde{c}_t \\ h_t &= o_t \odot \sigma_h(c_t) \end{aligned}$$



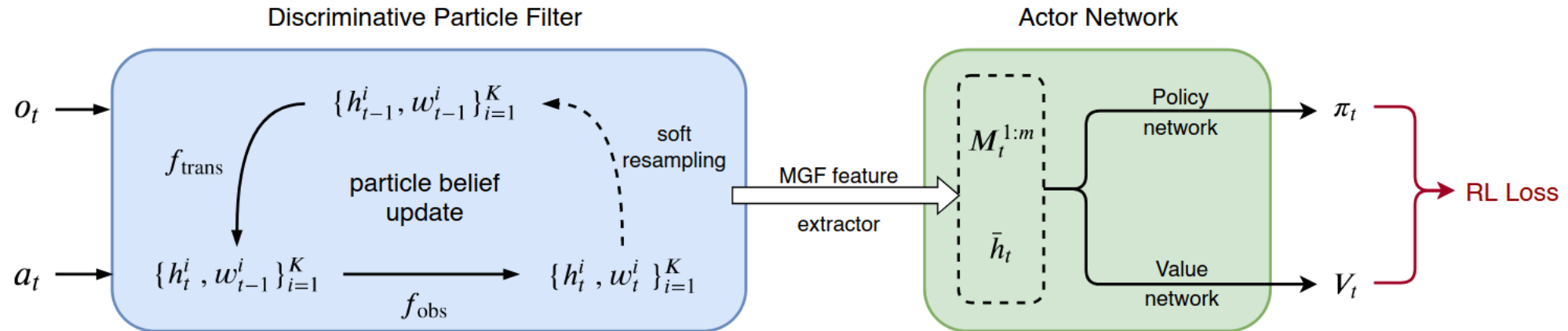
Legend:



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Approach 3: Particle Filters and MGF Features

Approach 3: Particle Filters and MGF Features



Published as a conference paper at ICLR 2020

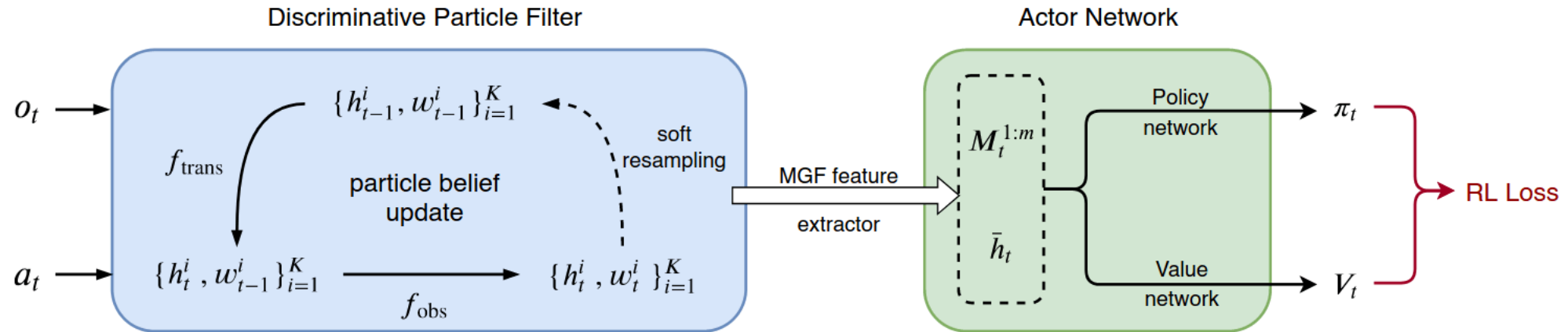
DISCRIMINATIVE PARTICLE FILTER REINFORCEMENT
LEARNING FOR COMPLEX PARTIAL OBSERVATIONS

Xiao Ma¹, Peter Karkus¹, David Hsu¹ and Wee Sun Lee¹, Nan Ye²

¹National University of Singapore, ²The University of Queensland

{xiao-ma, karkus, dyhsu, leews}@comp.nus.edu.sg, nan.ye@uq.edu.au

Approach 3: Particle Filters and MGF Features



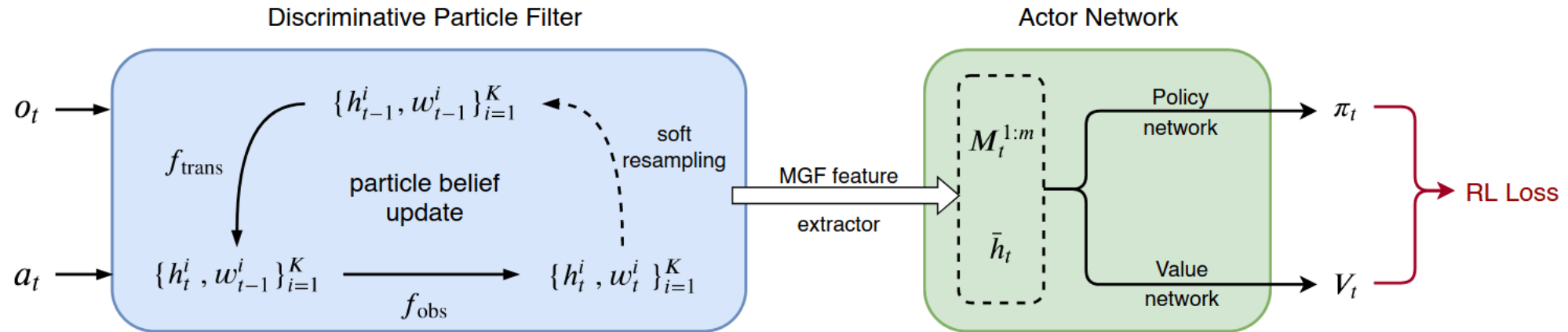
$$M_{\mathbf{X}}(\mathbf{v}) = \mathbb{E} \left[e^{\mathbf{v}^\top \mathbf{X}} \right]$$

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$$M_{\mathbf{X}}(\mathbf{v}) = \mathbb{E} \left[e^{\mathbf{v}^\top \mathbf{X}} \right]$$

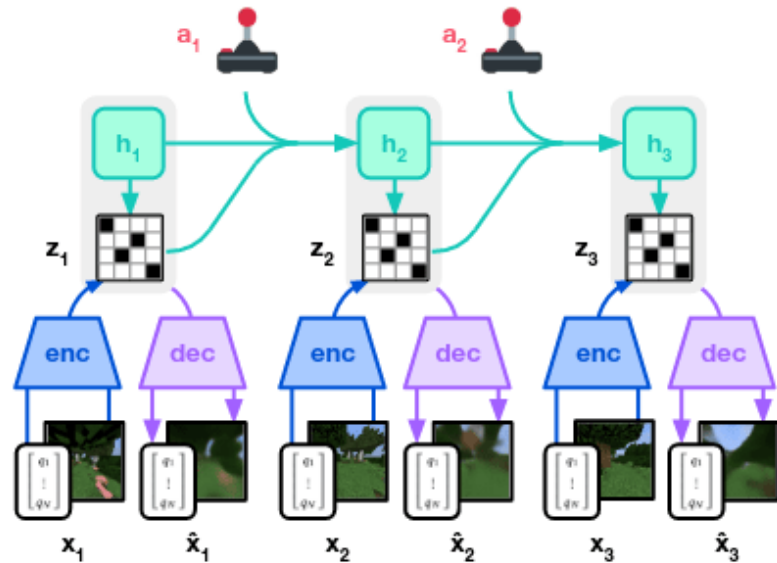
$$M_j = \frac{d^j M_{\mathbf{X}}}{d\mathbf{v}^j} \Big|_{\mathbf{v}=0}$$

Published as a conference paper at ICLR 2020

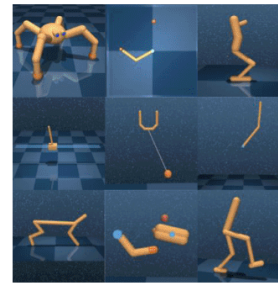
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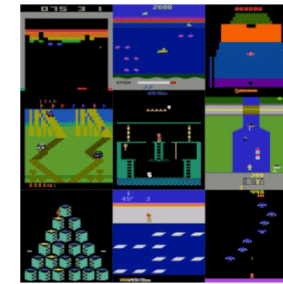
Approach 4: Latent Representations (e.g. World Models)



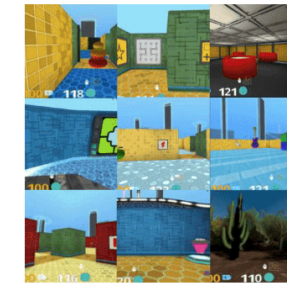
(a) World Model Learning



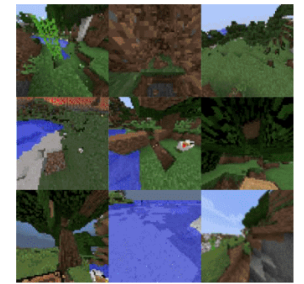
(a) Control Suite



(b) Atari



(c) DMLab



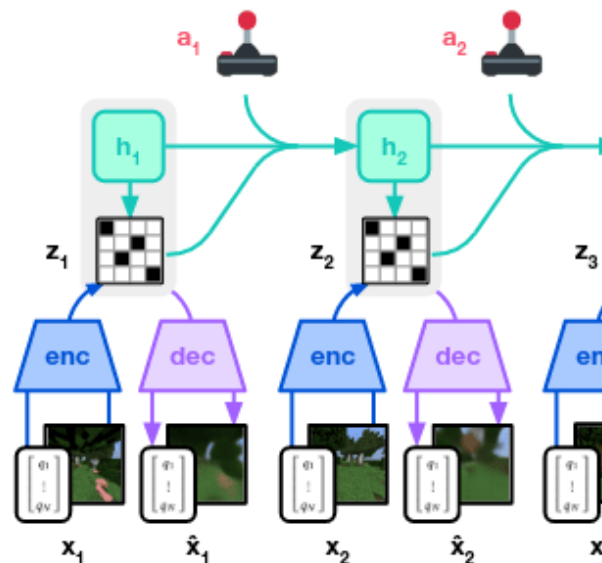
(d) Minecraft

Mastering Diverse Domains through World Models

Danijar Hafner,^{1,2} Jurgis Pasukonis,¹ Jimmy Ba,² Timothy Lillicrap¹

¹DeepMind ²University of Toronto

Approach



(a) World Model Learning

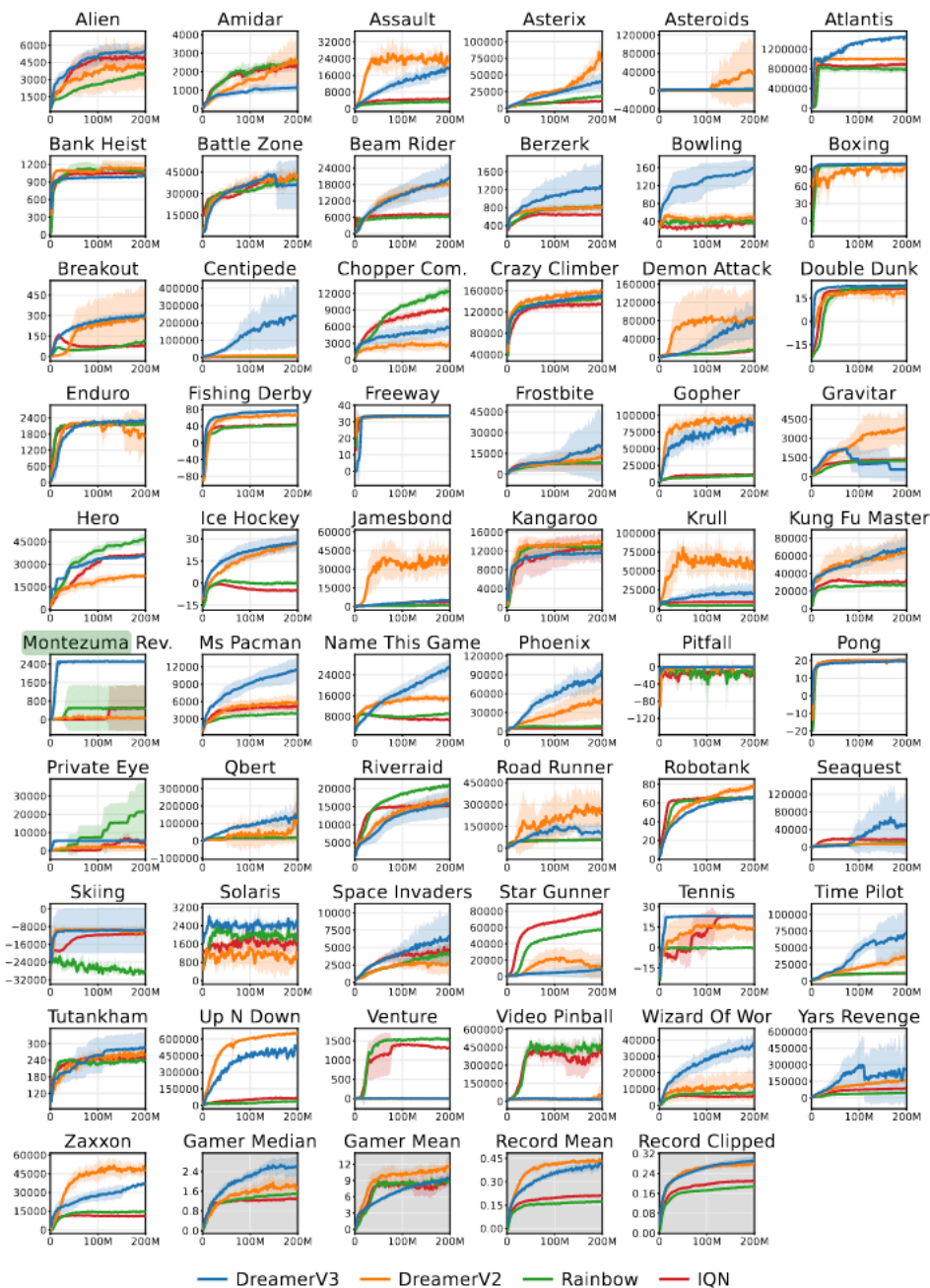
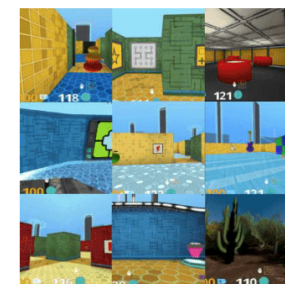


Figure U.1: Atari training curves with a budget of 200M frames.

Representations



(c) DMLab



(d) Minecraft

g Diverse Domains through World Models

Hafner,¹² Jurgis Pasukonis,¹ Jimmy Ba,² Timothy Lillicrap¹

¹DeepMind ²University of Toronto